Introduction to Remote Sensing in Agriculture

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(Course Outline P01)

I. Theme: Agriculture in the 21st C.

Section: Ag's Challenges and Solutions

1. Challenges
    (a) The Anthropocene
        i. Human Population Impacts
        ii. Human Economic Imapcts
    (b) Ecosystem Decline
    (c) Climate Change
    (d) Sustainability and Resiliency
2. Solutions
    (a) Agroecology and Regenerative Ag
    (b) Urban Ag
    (c) Tribal Knowledge
    (d) Advanced Crop Breeding
    (e) Adaptive Management
    (f) Tech
        i. Precision Ag
        ii. Hydroponics
        iii.

Section: Remote Sensing Tools

1. Satellites
    (a) Sentinel-2
    (b) Landsat 8

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   (c) MODIS

2. Drones
   (a) Reconnaissance Drones
   (b) Crop Management Drones
3. Data and Maps
   (a) Data Acquistion
      i. Data Analytics
   (b) Maps

II. Theme: Cloud Computing

Section: Professional AI and API Use

1. Cloud, API, and AI Service Providers
   (a) Anthropic
   (b) Google
      i. Google Cloud
      ii. Google Earth Engine
   (c) Copernicus
2. Accessing AIs and APIs
   (a) Python
   (b) Spyder
   (c) SDKs

III. Theme: Records Management

Records and Document Management

1. QAQC for Docs
2. GitHub

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IV. NOTES

Section: Ag's Challenges and Solutions

Challenges

- In current and future climate scenarios the resilience and productivity of agricultural systems will be increasingly jeopardized. Furthermore, population growth trends, expected to reach 8.7 billion by 2030 and 9.7 billion by 2050, will further strain food production systems worldwide, which so far have not been able to keep pace (**Segarra et al., 2020**)
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Solutions

- Management-derived challenges as well as abiotic and biotic stressors may all be confronted and their negative impacts greatly ameliorated by employing remote sensing technologies to better guide adaptive agricultural management decisions, such as variable rate applications and treatments, which can be assessed and confronted at different levels of precision using different sources of remotely sensed data.

Section: Introduction to Remote Sensing in Ag

Satellites

- remote sensing using multispectral images is a proxy for extensive manual crop monitoring operations in the field and provides potential savings in precious time and resources. Precision agriculture has grown and established itself as a central tool in agricultural management to forecast yield, monitor plant stress, and optimize fertilization, irrigation, and soil tilling activities for a more sustainable management of farming practices and improvements in yield for farmers.(**Segarra et al., 2020**)
- Sentinel-2 platform is the pivotal sensor: by providing freely accessible multispectral imagery with 10–20 m spatial resolution, frequent revisit times, and red-edge bands, Sentinel-2 has enabled a shift from regional assessments toward field-level and within-field yield monitoring, and its red-edge bands are also effective for monitoring nitrogen status. The Sentinel-2 constellation, with improved spatial, spectral, and temporal resolution was designed specifically to meet the needs of the agricultural community, both farmers and academic researchers with a focus on international agricultural development. (**Segarra et al., 2020**)
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Satellite Data

- Concerning agriculture, in situations where crops interact with any given aspect of their environment (seasonal climatic variations, meteorological extreme events, pests, soil properties, etc.) or as crops grow and pass through different phenological stages, the interactions between plants and light reflectance translate into changes in plant signal patterns that can be interpreted using satellite data.(**Segarra et al., 2020**)

- One very notable use of Sentinel-2 features at field monitoring is the ESA Sen2-Agri system which generates modules of time series along the growing season. These different products consist of a monthly cloud-free composite of surface reflectance at 10–20 m resolution; monthly dynamic cropland masks, delivered from the agricultural mid-season onwards; a cultivated crop type map at 10 m resolution for main crop groups; and time series of vegetation status indicators, NDVI and LAI (**Segarra et al., 2020**)

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References

Segarra, J., Buchaillot, M. L., Araus, J. L., & Kefauver, S. C. (2020). Remote Sensing for Precision Agriculture: Sentinel-2 Improved Features and Applications. *Agronomy*, *10*(5) (cit. on pp. 3, 4).

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